Md Mashuk E Lahi

Software, Data & Embedded Systems Engineer (entry level) | Master of Software Engineering

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Full NZ work rights | Available immediately | Open to relocate anywhere in NZ

PROFESSIONAL SUMMARY

Entry-level engineer with a Master of Software Engineering (A average) from Yoobee College, Auckland, and eight years of hands-on work across embedded hardware, IoT, testing and field support in Bangladesh. At DataSoft Systems I took connected devices from circuit and PCB design through firmware to field rollout, led a team of seven on a 450-unit deployment, and received the Best IoT Engineer award in 2023. In my Master's I built a live full-stack marketplace, a Power BI analytics study and a PyTorch forecasting model. I pick up new tools quickly and work well between hardware, software and non-technical teams, and I am looking for a graduate or junior role where I can learn fast and be useful early.

KEY SKILLS

Software: Python, JavaScript, C/C++; Django REST Framework, React, PostgreSQL; Docker, Git and GitHub, pytest; Agile

Data and machine learning: Power BI, Tableau, Excel; Pandas, scikit-learn, PyTorch; statistics, regression, time-series forecasting

Embedded and IoT: ESP32, Arduino, FreeRTOS; MQTT, CoAP, LoRa, GSM, Wi-Fi, BLE; UART, SPI, I2C; PCB design (Eagle, KiCad)

Testing and support: Firmware testing, accuracy validation against reference instruments, defect reporting, Jira; field troubleshooting, manuals and user training

Project delivery: Leading a team of seven, hardware sourcing and procurement, client handover

Languages: English (fluent), Bangla (native), Hindi (basic), Urdu (basic)

PROFESSIONAL EXPERIENCE

IoT & Embedded Systems Engineer | DataSoft Systems Bangladesh Ltd., Dhaka Nov 2016 - Oct 2025

Joined as IoT Trainee (Nov 2016); Software Developer 2017, IoT Engineer 2018, Senior Software Engineer 2022

- Led a team of seven to deliver 450 smart sockets that monitored and remotely switched a client's air purifiers; sourced hardware overseas, validated every unit with firmware version tracking and anomaly logs, and caught 15 faulty devices.
- Architected the device side of a building security and energy monitoring system: an RF hub for door, motion and gas sensors and a CT clamp board for floor-by-floor electricity use, with circuit, PCB and firmware design and data sent over MQTT.
- Built sensing nodes for a World Bank-funded air quality network of 40+ nodes producing over 3.4 million readings a day, with sensors calibrated to within 7 to 9% of reference instruments.
- Managed an IoT cattle shed monitoring system end to end over three to four months, from field inspection to commissioning automated fans, misting and lighting. This project and the smart socket rollout together earned the Best IoT Engineer award (2023).
- Led the hardware team for a government bus validator and POS rollout in Dhaka, from device selection and procurement to a year of support; restored service within one to two hours per fault and trained 40 government staff.
- Delivered smart home automation for clients from device selection to handover, with Google Home and Alexa voice control.
- Designed electricity and water consumption sensors for the Department of Telecommunications, including firmware testing and installation guides.

Part-time work alongside study | Auckland Dec 2025 - Present

Warehouse Operations Associate, My Food Bag (Apr 2026 - present); Housekeeping Attendant, Rose Park Hotel, Parnell (Dec 2025 - Apr 2026)

PROJECTS

SureSource: B2B Farm-to-Business Marketplace | Master's coursework, sole developer 2026

Django REST Framework, React, PostgreSQL, Docker, pytest | suresource.vercel.app

- Designed, built and deployed a live multi-role marketplace on my own, from REST API and React frontend to Docker, cloud deployment and automated tests.

Fitness App User Engagement Analysis | Master's coursework (Data Analytics) 2026

Python, Pandas, SciPy, scikit-learn, Power BI | github.com/Mashuk25/DataAnalytics

- Analysed 5,000 user records with hypothesis tests, regression and classification, and presented the findings in Power BI dashboards for non-technical readers.

LoadCast: Household Electricity Demand Forecasting | Master's capstone 2026

Python, PyTorch, PyTorch Forecasting, Pandas | github.com/Mashuk25/LoadCast

- Built a Temporal Fusion Transformer that caught 13% more household demand peaks than an LSTM baseline (three runs, significance-tested).

EDUCATION

Master of Software Engineering | Yoobee College of Creative Innovation, Auckland 2025 - 2026

Grade average: A. Relevant courses: Professional Software Engineering, Data Analytics, Cloud Security, Intelligent Transportation Systems, industry-based capstone.

B.Sc. Electronics & Telecommunications Engineering | East West University, Bangladesh 2012 - 2016

GPA 3.18 / 4.0

AWARDS AND TRAINING

- Best IoT Engineer Award, DataSoft Systems Bangladesh (2023)
- IoT Army of 300 (Batch 1), national IoT training programme run by the ICT Division, BUET, ULAB and DataSoft (2016 - 2017)
- Power & Energy Hackathon, Ministry of Power, Energy & Mineral Resources, Bangladesh (2017)

References available on request.